

NN Introduction

2023-10-20

Confidential!
nick@khadas.com
2024-11-20 08:49:12

Contents



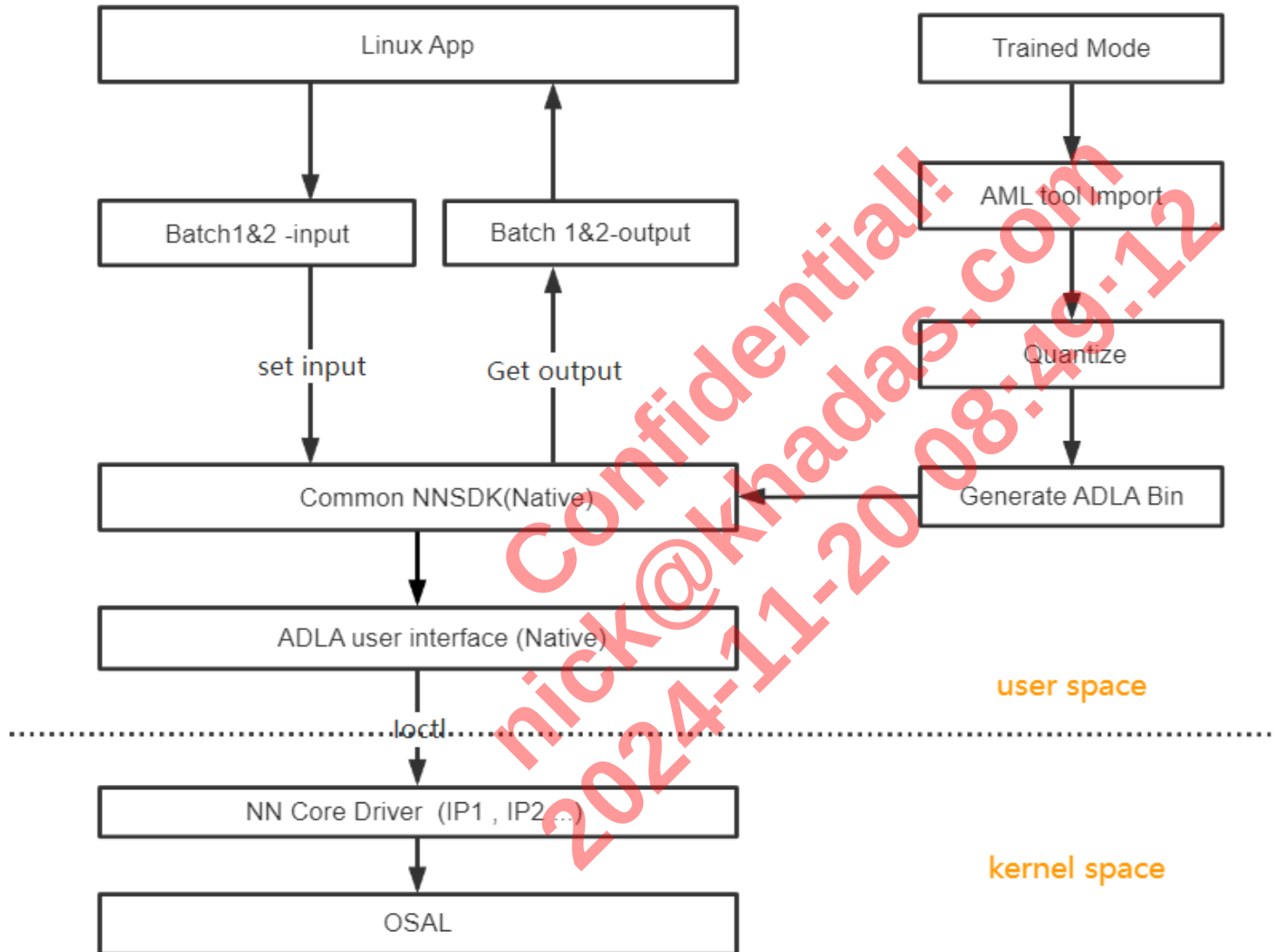
- Amlogic NN Summary
- NN Software Flow
- NN Software Stack
- Model Convert
- NN OP Support List
- NN Performance and Bandwidth
- NN Accuracy
- Amlogic NN Release Source

Confidential!
nick@khadas.com
2024-11-20 08:49:12

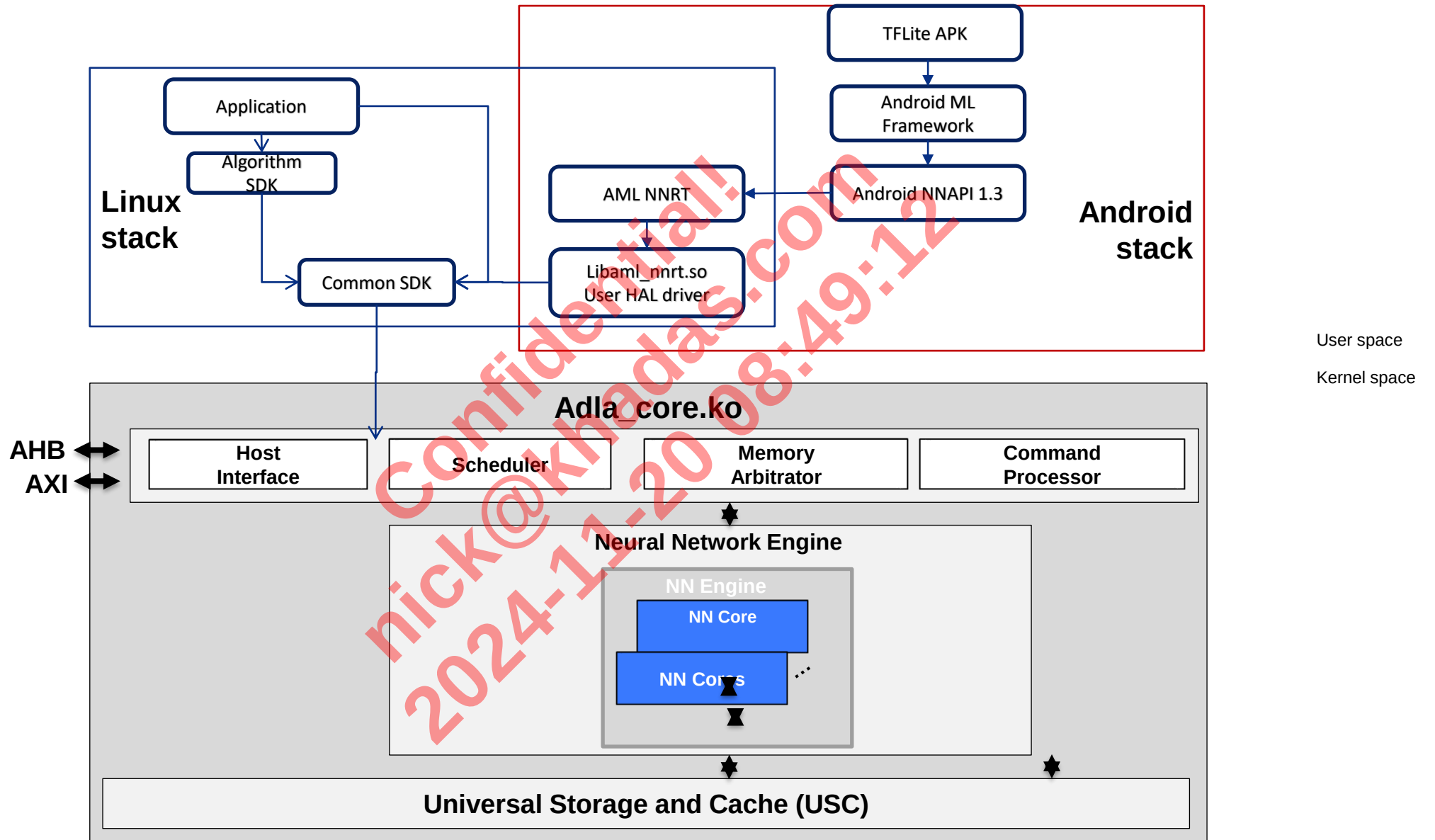
Amlogic NN Summary

- Amlogic ADLA is a neural acceleration engine. It is designed to accelerate the inference of AI image models such as image classification, detection and segmentation. It can also be used for AI voice models such as speech to text, text to speech, voice denoise, keyword detection.
- Supports caffe, tensorflow, tflite, darknet, onnx, Keras, Pytorch, Paddle, Mxnet open-source models
- Supports three quantization methods including int8, int16, uint8, etc.
- The supported open-source framework model needs to be quantified using the above quantification method first, and then convert to adla file that the chip can recognize and load
- Supports Android, Linux, FreeRTOS

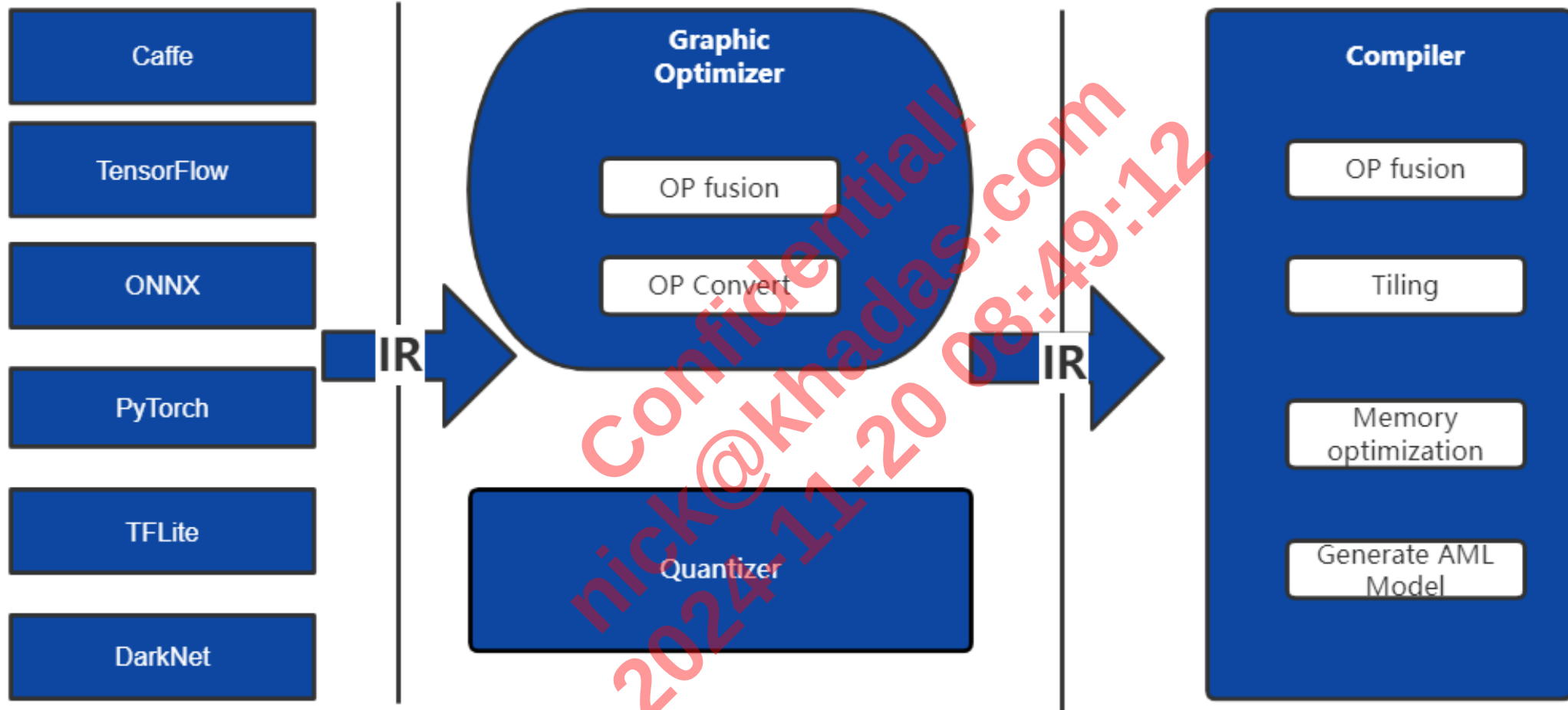
NN Software Flow



NN Software Stack



Model Convert



NN OP Support List

tflite op	tflite op	tflite op	tflite op	tflite op	tflite op	tflite op	tflite op	tflite op
ABS	DEPTH_TO_SPACE	LSTM	NEG	REDUCE_MIN	SLICE	ARGMAX	L2_NORMALIZATION	SIN
ADD	EMBEDDING_LOOKUP	LESS	NOT_EQUAL	REDUCE_ANY	SUM	ARG_MIN	LOCAL_RESPONSE_NORMALIZATION	SQRT
ADDN	EXPAND_DIMS	LESS_EQUAL	PAD	RESIZE_NEAREST_NEIGHBOR	SHAPE	CAST	LOG_SOFTMAX	SCATTER_ND
AVERAGE_POOL_2D	EQUAL	LOGICAL_OR	PRELU	REVERSE_V2	SQUARE	CEIL	LOG	TOPK_V2
BATCH_TO_SPACE_ND	FULLY_CONNECTED	LOGICAL_AND	PADV2	REDUCE_ALL	SQUARED_DIFFERENCE	COS	NON_MAX_SUPPRESSION_V4	UNIQUE
BIDIRECTIONAL_SEQUENCE_LSTM	FILL	LOGICAL_NOT	PACK	RNN	SPLIT_V	DEQUANTIZE	NON_MAX_SUPPRESSION_V5	MULTINOMIAL
BATCH_MATMUL	GATHER	LEAKY_RELU	QUANTIZE	SPACE_TO_DEPTH	SELECT_V2	DIV	ONE_HOT	
BROADCAST_TO	GREATER	MAX_POOL_2D	RELU	SPACE_TO_BATCH_ND	TANH	EXP	POW	
BROADCAST_ARGS	GREATER_EQUAL	MUL	RELU_N1_TO_1	SUB	TRANSPOSE	ELU	RSQRT	
CONCATENATION	GATHER_ND	MEAN	RELU6	SQUEEZE	TRANSPOSE_CONV	FLOOR	REDUCE_PROD	
CONV_2D	HARD_SWISH	MAXIMUM	RESHAPE	STRIDED_SLICE	TILE	FLOOR_DIV	RANGE	
CUMSUM	L2_POOL_2D	MINIMUM	RESIZE_BILINEAR	SPLIT	UNIDIRECTIONAL_SEQUENCE_LSTM	FLOOR_MOD	ROUND	
DEPTHWISE_CONV_2D	LOGISTIC	MIRROR_PAD	REDUCE_MAX	SELECT	UNPACK	GELU	SOFTMAX	

NN Performance and Bandwidth

		Board: C308L			Board:T7C	
Network	Dtype	Input size	FPS	Bandwidth(MB/inference)	FPS	Bandwidth(MB/inference)
resnet50	int8	224*224*3	100.17	35.94	219.3	28.76
mobilenet		224*224*3	561.48	3.92	987.17	4.17
inception_v1		224*224*3	214.78	7.55	518.94	6.55
inception_v2		224*224*3	182.32	11.25	390.17	10.81
inception_v3		299*299*3	65.82	28.92	163.85	23.22
inception_v4		299*299*3	32.35	72.6	86.17	51.39
mobilenet_v1		224*224*3	564.33	4.14	986.19	4.17
resnet_v1_50		224*224*3	105.76	33.84	230.04	27.53
resnet_v1_101		224*224*3	59.78	53.7	149.52	45.6
resnet_v1_152		224*224*3	39.96	74.25	108.26	61.53
resnet_v2_50		299*299*3	54.32	59.95	115.53	45.85
resnet_v2_50		224*224*3	100.53	34.21	207.21	27.75
resnet_v2_101		224*224*3	56.59	54.65	129.38	45.81
resnet_v2_152		224*224*3	38.97	76.57	91.83	62.54
vgg_16		224*224*3	20.17	146.88	44.93	149.29
retinaface		640*384*3	251.57	4.27	347.95	2.08
SSD-mobilnet		300*300*3	162.58	14.81	252.02	17
mobilenetv2		224*224*3	657.46	3.53	802.57	3.47
yolov2		416*416*3	20.4	75.98	66.66	73.96

NN Accuracy

FrameWork	Mode_Name	Input_size	ADLA Accuracy Status(5000 imagenet dataset)								Board Accuracy on C3(5000 imagenet dataset)			
			Original Model		Int8 Perchannel		Int8 Perlayer		Int16		Int8 Perchannel		Drop	
			Top1	Top5	Top1	Top5	Top1	Top5	Top1	Top5	Top1	Top5	Top1	Top5
PYTORCH	resnet_18	[1,3,224,224]	63.02%	84.78%	62.90%	84.64%	63.14%	84.42%	63.02%	84.72%	62.58%	84.58%	0.32%	0.06%
	resnet_34	[1,3,224,224]	67.70%	88.24%	67.94%	88.24%	67.70%	88.12%	68.14%	88.72%	68.08%	88.36%	-0.14%	-0.12%
	resnet_50	[1,3,224,224]	71.36%	90.12%	71.50%	90.20%	71.34%	90.08%	71.44%	90.22%	71.32%	90.18%	0.18%	0.02%
	resnet_v1_101	[1,3,224,224]	72.84%	91.00%	69.26%	88.98%	68.50%	88.42%	72.76%	91.12%	69.2%	89.38%	0.06%	-0.4%
	squeezenet 1.0	[1,3,224,224]	43.14%	69.16%	43.16%	68.78%	41.68%	67.58%	44.78%	69.74%	43.3%	68.5%	0.14%	0.28%
	SqueezeNet 1.1	[1,3,224,224]	44.38%	69.70%	43.06%	68.32%	41.82%	67.94%	45.14%	69.88%	43.1%	68.5%	-0.04%	-0.18%
	resnext50_32x4d	[1, 3, 300, 300]	72.54%	91.14%	72.86%	91.18%	74.24%	91.16%	73.78%	91.64%	73.45%	91.25%	-0.59%	-0.07%
ONNX	resnet18-v1-7	[1, 3, 224, 224]	66.14%	87.92%	66.26%	87.80%	64.36%	87.12%	66.40%	87.80%	66.48%	87.72%	-0.22%	0.08%
	resnet18-v2-7	[1, 3, 224, 224]	66.70%	87.96%	62.16%	84.26%	64.80%	85.86%	66.68%	87.78%	62.5%	84.54%	0.34%	-0.28%
	resnet34-v1-7	[1, 3, 224, 224]	71.48%	90.16%	71.18%	90.22%	70.56%	89.54%	71.54%	90.34%	71.28%	90.16%	-0.10%	0.06%
	resnet34-v2-7	[1, 3, 224, 224]	71.46%	90.66%	71.58%	90.20%	70.00%	89.56%	71.66%	90.52%	71.48%	90.14%	0.10%	0.06%
	resnet50-v1-7	[1, 3, 224, 224]	72.32%	90.88%	71.60%	90.66%	70.56%	90.16%	72.20%	90.80%	71.18%	90.52%	0.42%	0.14%
	resnet50-v2-7	[1, 3, 224, 224]	73.74%	91.08%	70.96%	89.98%	70.20%	88.64%	73.58%	91.00%	70.90%	89.88%	0.06%	0.10%
	squeezenet1.1-7	[1, 3, 224, 224]	52.00%	75.96%	49.10%	73.70%	48.10%	72.98%	51.40%	75.58%	49.68%	73.52%	-0.58%	0.18%
	vgg16-7	[1, 3, 224, 224]	69.30%	89.48%	71.58%	90.08%	69.34%	89.26%	70.50%	89.96%	70.36%	90.00%	1.22%	0.08%
	vgg19-7	[1, 3, 224, 224]	71.10%	90.74%	71.78%	91.04%	70.92%	90.32%	71.86%	91.00%	71.78%	91.04%	0.00%	0.00%
	vgg19-bn-7	[1, 3, 224, 224]	70.58%	90.08%	71.46%	90.82%	69.62%	89.88%	71.72%	90.86%	71.80%	90.82%	-0.34%	0.00%
CAFFE	inception-v3	[1,3,299,299]	75.94%	92.90%	76.08%	92.68%	75.64%	92.54%	76.02%	93.02%	75.88%	93.13%	0.21%	-0.45%
	inception-v4	[1,3,299,299]	77.28%	93.58%	72.56%	90.18%	72.80%	90.60%	76.88%	93.44%	72.28%	90.83%	0.29%	-0.64%

Amlogic NN Release Source

Document Title	Note
Amlogic_NN_Introduction	Overall introduction
Amlogic_NN_Model_Design_guide	Model design guide
Neural_Network_layer_and_Operation_Support_list	Supported layers and operation (Todo)
Model Conversion User Guide (1.x)	Model conversion guide
adla-toolkit-binary-x.xx.xx.tgz	Model Conversion tool(Contains sdk development kit)
ADLA_SDK API (1.x.x)_CN	Amlogic sdk interface document
amlogic_adla_tool_docker_release.tar.gz	The docker file that has been configured to compile and convert the environment

Amlogic NN Release Source

How to obtain tools:

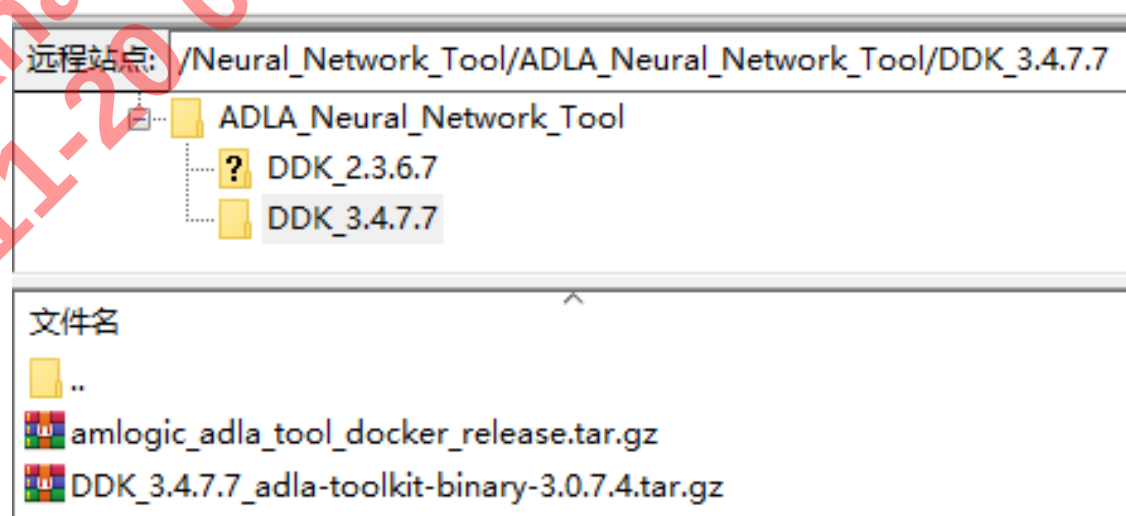
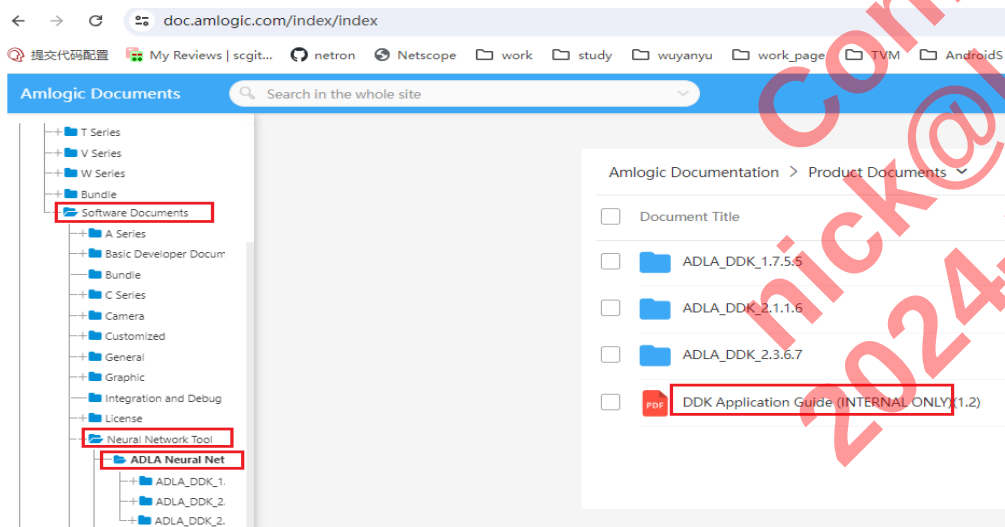
FTP server: ftp-china.amlogic.com

account: amlogic_nn_tool_release.guest

FTP server Path: /Neural_Network_Tool/ADLA_Neural_Network_Tool/

adla_tool is released through ftp sever. Customers please contact the amlogic docking person to obtain the latest access password from the DDK Application Guide (INTERNAL ONLY).doc in the application document section and provide it to the customer. The port defaults or is set to 21.

Link: <https://doc.amlogic.com/index/index>



THANK YOU!

Confidential!
nick@khatas.com
2024-11-20 00:49:12